Ali Husain

Software engineer and product designer

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Summary

I build complete systems, from the interface to the infrastructure.

- 11 public repositories, each one specified, tested, documented and released
- 4,255 automated tests across them, run on every change
- 15 services in production on a platform I designed and operate

Projects

Ballast Live app, open source · gallowforge.github.io/ballast · github.com/gallowforge/ballast
Personal finance that turns savings into a salary that lasts. Designed and built end to end, offline-first, with no account and no server. Says what is safe to spend today, draws how long the money lasts, and reads bank statements on the device, checking every row against the running balance.
599 automated tests, unit and end-to-end. 40 currencies, in exact whole-number maths.

Ethereal Live app, open source · gallowforge.github.io/Ethereal · github.com/gallowforge/Ethereal
A calm coach for training, food and the shape of the day. An installable app that plans the day, guides workouts and tracks nutrition, with no account, no ads and no trackers. Product design, front end and delivery, end to end, built to a WCAG 2.1 AA target with Lighthouse at 100.
397 automated tests, run on every push. 1,474 foods in a searchable offline database.

Holdfast Live demo, open source · gallowforge.github.io/holdfast · github.com/gallowforge/holdfast
A memory for coding agents in which every fact carries its own proof. Each fact is stored with a check that proves it, run again before the agent is told. What has stopped being true comes back marked broken, with the evidence. One Go binary.
1,651 automated tests, on Linux and macOS. 10 kinds of check, from a file to a live endpoint.

Seaward Live demo, open source · gallowforge.github.io/seaward · github.com/gallowforge/seaward
Shows which ports on a server can really be reached from outside, and the rule that decides each one. Reads the rules the kernel has loaded and follows a packet through them, so a port Docker published past the firewall's list is found and explained. One Go binary.
964 automated tests, run on every change. 1,950 real packets checked against the kernel.

Vesper In production, open source · gallowforge.github.io/vesper · github.com/gallowforge/vesper
A read-only operations console that answers one question: does anything need you? One Go binary with the interface inside, read-only by construction. When it cannot confirm something it says so instead of showing an all-clear. Now at version 2.2.0.
7.4 MB: one Go binary, about 27 MB of memory. 191 tests, 34 end to end under real systemd.

homelab Designer and sole operator · github.com/gallowforge/homelab
Fifteen services on one small machine, run as code, with backups that are proven every week. Remote access with zero port forwards. Every backup set is verified nightly, copied to three machines and restored in a weekly drill.
410 commits, about 4,100 lines of runbooks.
15 containers, with 21 scheduled services. 77 alert conditions and 13 uptime probes.

Tools

agent-guardrails (github.com/gallowforge/agent-guardrails): Reviewer and verifier agents that judge work they did not write, and a hook that warns a coding agent before a destructive command runs. 36 tests.

evalbench (github.com/gallowforge/evalbench): Tells you which language model to use for your own job: the cheapest one that clears your bar, proved by automatic checks and blind ratings. 291 tests.

Skills

Product design. Interfaces designed for phone and desktop together, accessible by default, and measured with real devices and automated audits. Seen in Ballast, Ethereal, Vesper: Ethereal to a WCAG 2.1 AA target, with Lighthouse at 100.

Front end. Installable apps that work offline, with no account and no server behind them, and a live operations console streamed to the browser. Seen in Ballast, Ethereal, Vesper: 599 and 397 automated tests on the two apps.

Back end. Services and command-line tools in Python, Go, TypeScript, Bash, SQL and C/C++, with the documentation that ships with them. Seen in Holdfast, Seaward, Vesper, Evalbench: three Go binaries, each with no third-party Go dependencies.

AI evaluation and agent workflows. Model evaluation with automatic checks and blind human rating. Agent memory that is proved again before use, and reviewer and verifier agents that judge work they did not write. Seen in Holdfast, Evalbench, agent-guardrails: one build ran as nine coordinated agents, a platform audit as thirteen.

Infrastructure and operations. Linux, containers, systemd, CI/CD and infrastructure as code, with monitoring and incident response. Backups are proven by a restore drill every week. Seen in homelab, Vesper: 410 commits, 21 scheduled services, about 4,100 lines of runbooks.

Security practice. DNS, TLS, mesh VPN, firewalls, service sandboxing, intrusion prevention, encryption and secrets handling. Zero port forwards and one audited public endpoint. Seen in Seaward, homelab, Vesper, Evalbench: 7-node mesh VPN; code tests run in a sandbox with no network.

Video pipeline. A documented production method for painted films made with AI image and video models, with post-production built in code, from beat-matched editing to 4K mastering on GPU. Seen in filmforge: 9 skills and 7 tested tools; every encode verified.

How I work

Specified before it is built, tested before it ships, documented so it runs without me.

Prove it works. A backup isn't done until it's restored; a fix isn't done until it's tested from the user's side.

Write it down. Every system ships with a runbook someone else could follow.

Learn by shipping. Linux, Go, AI agents, post-production: each learned by building something real.

Fewer things, done properly. Simple setups that stay up beat clever ones that break.

Automated tests by repository. holdfast 1,651, seaward 964, ballast 599, Ethereal 397, evalbench 291, vesper 191, homelab 43, filmforge 40, agent-guardrails 36, aih 31, bwtidy 12. Total 4,255 across 11 public repositories.

Education

Computer Science and Engineering

Contact

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